School of Electrical Engineering and Computer Science

Advanced Multi-Agent Threat Analysis System: Explainable Network Intrusion Detection through LLM-Based Multi-Agent Debate

by

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Prof Michael Bruenig
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Dear Professor Bruenig,

In accordance with the requirements of the Degree of Bachelor of Engineering (Honours) in the School of Electrical Engineering and Computer Science, I submit the following thesis entitled "Advanced Multi-Agent Threat Analysis System: Explainable Network Intrusion Detection through LLM-Based Multi-Agent Debate". The thesis was performed under the supervision of Dr Siamak Layeghy and Dr Marius Portmann. I declare that the work submitted in the thesis is my own, except as acknowledged in the text and footnotes, and that it has not previously been submitted for a degree at the University of Queensland or any other institution.

Yours sincerely,

Kunwar Singh

Declaration by Author

This thesis is composed of my original work, and contains no material previously published, written by another person, or generated by artificial intelligence (AI), except where clearly referenced and acknowledged either in the main text or the preliminary pages of this thesis. I have clearly stated the contribution of others to jointly authored works that I have included in my thesis.

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Use of AI Statement

I acknowledge the use of generative AI tools in completing this thesis. Details of which tools were used and how they were used are provided in the table below, along with appropriate in-text and full references. I take responsibility for critically evaluating and integrating the AI-generated content, and ensuring it adheres to academic integrity standards.

How AI was used in this project:

- **Topic exploration:** Yes. Claude (Anthropic) and ChatGPT (OpenAI) were used to explore the landscape of LLM applications in cybersecurity, identify relevant sub-areas, and generate initial keywords for literature searches. All identified topics were independently verified through manual database searches.
- **Research question definition:** No. Research questions were defined through supervisor meetings, iterative refinement based on experimental results, and manual review of the literature. The pivot from MCP-centric to multi-agent architecture was driven by empirical findings, not AI suggestions.
- **Literature review:** Yes. AI assistants were used to summarise papers already located through manual database searches and to suggest thematic groupings. All references were independently verified against the original papers, and all claims attributed to cited works were checked for accuracy.
- **Figure generation:** No. All figures, diagrams, and charts were created manually using Python (matplotlib), TikZ, or the React-based dashboard.
- **Data analysis:** Yes. AI assistants (Claude Code) were used to write Python scripts for metrics calculation, result aggregation, and cost analysis. All scripts were reviewed, and results were independently verified against raw data.
- **Data generation:** No. All data comes from the CICIDS2018 NetFlow v3 dataset, a publicly available benchmark. No synthetic data was generated.
- **Text generation:** Yes. AI assistants (Claude) were used to draft sections of this thesis. All AI-generated text was critically reviewed, edited for accuracy and

argument coherence, fact-checked against experimental results and cited sources, and revised to reflect the author's own analysis and conclusions. The intellectual structure, arguments, and interpretation of results are the author's own work.

- **Formatting:** Yes. AI assistants were used to convert content to LaTeX format, resolve compilation issues, and ensure consistent formatting. The intellectual content, structure, and arguments are the author's own.

- **Code generation:** Yes. AI assistants (Claude Code, GitHub Copilot) were used extensively to write the AMATAS system code, including the multi-agent pipeline, MCP server, dashboard, and evaluation scripts. All code was reviewed, tested, and debugged by the author. The architectural decisions (the six-agent design, Devil's Advocate mechanism, two-tier pre-filter, and consensus weighting) were made by the author based on experimental evidence and supervisor guidance.

AI Model & date	Lang. Transl.	Grammar/style	Topic expl.	Research Qs	Lit. review	Data analysis	Gen. data	Gen. text	Visual content	Code	Feedback	Other
Claude 3.5/4 (Anthropic) Jul '25–Apr '26		✓	✓		✓	✓		✓		✓	✓	
GPT-4o (OpenAI) Jul '25–Apr '26			✓									AMATAS agent
GitHub Copilot Jul '25–Apr '26										✓		

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Abstract

This thesis presents AMATAS (Advanced Multi-Agent Threat Analysis System), a network intrusion detection architecture that combines a machine learning pre-filter with a six-agent large language model pipeline to deliver both accurate detection and transparent reasoning. Unlike traditional classifiers that produce opaque probability outputs, AMATAS generates multi-perspective reasoning chains from specialist agents analysing protocol conformance, statistical anomalies, behavioural signatures, and temporal patterns, with an adversarial Devil's Advocate mechanism for false positive control.

Evaluated across all fourteen attack types in the CICIDS2018 NetFlow v3 dataset at realistic class distributions (five per cent attack prevalence), AMATAS achieves a mean F1 score of 92.9 per cent across thirteen detectable attack categories, with a false positive rate of zero per cent in ten of fourteen experiments. A two-tier architecture combining a Random Forest pre-filter with the LLM pipeline reduces analysis cost by 94.6 per cent, from an estimated $509.78 to $27.35 for the complete evaluation corpus of 14,000 flows.

A controlled ablation study demonstrates that the multi-agent architecture outperforms the best single-agent configuration by 25.2 F1 percentage points, and that external threat intelligence tools provide no measurable benefit on anonymised datasets. The Infiltration attack category produces zero per cent recall under the v2 per-flow architecture, exposing a fundamental limitation of individual flow-level analysis. A v3 temporal clustering experiment recovers 58% recall on Infiltration by presenting agents with multi-flow clusters grouped by source IP, though at the cost of an elevated 19.2% false positive rate that highlights the inherent difficulty of detecting multi-stage intrusion campaigns.

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